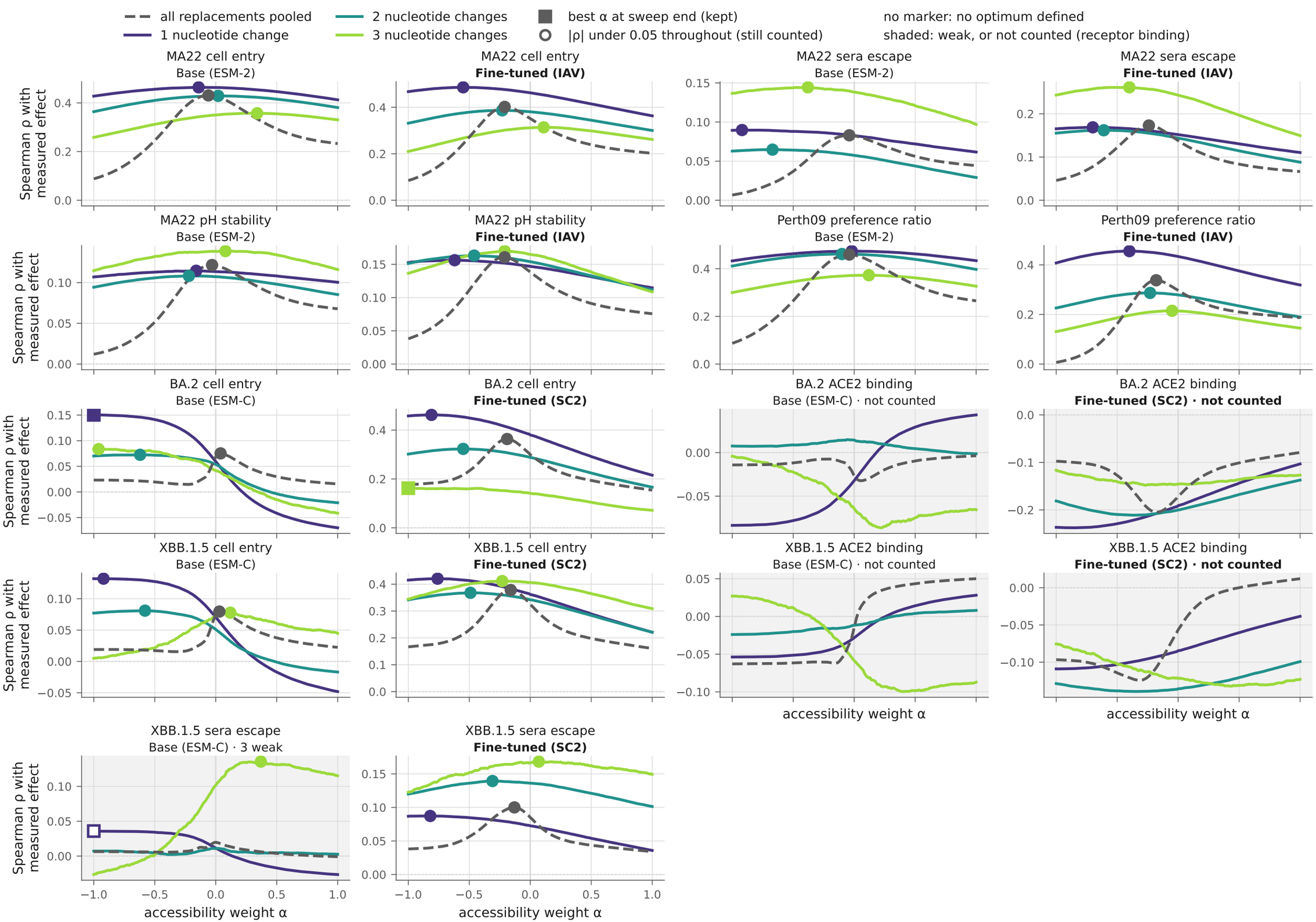

all replacements pooled
1 nucleotide change
2 nucleotide changes
3 nucleotide changes
best α at sweep end (kept)
|ρ| under 0.05 throughout (still counted)
no marker: no optimum defined
shaded: weak, or not counted (receptor binding)
MA22 cell entry
Base (ESM-2)
MA22 cell entry
Fine-tuned (IAV)
MA22 sera escape
Base (ESM-2)
MA22 sera escape
Fine-tuned (IAV)
MA22 pH stability
Base (ESM-2)
MA22 pH stability
Fine-tuned (IAV)
Perth09 preference ratio
Base (ESM-2)
Perth09 preference ratio
Fine-tuned (IAV)
BA.2 cell entry
Base (ESM-C)
BA.2 cell entry
Fine-tuned (SC2)
BA.2 ACE2 binding
Base (ESM-C) · not counted
BA.2 ACE2 binding
Fine-tuned (SC2) · not counted
XBB.1.5 cell entry
Base (ESM-C)
XBB.1.5 cell entry
Fine-tuned (SC2)
XBB.1.5 ACE2 binding
Base (ESM-C) · not counted
XBB.1.5 ACE2 binding
Fine-tuned (SC2) · not counted
XBB.1.5 sera escape
Base (ESM-C) · 3 weak
XBB.1.5 sera escape
Fine-tuned (SC2)
Spearman ρ with measured effect
accessibility weight α